

Notes: Tobin (RES, 1958)

Liquidity Preference as Behavior Towards Risk

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	This is a theory paper: what the “key objects” are	4
2.2	Transactions balances versus investment balances	4
2.3	Assets, returns, and risk objects	4
2.4	Investor types, preferences, and timing	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Keynesian benchmark: inelastic expectations of future interest rates	5
3.2	Aggregation under sticky expectations	6
3.3	The risk-aversion foundation: portfolio return, expected return, and risk	6
3.4	Preferences over expected return and risk	7
3.5	First rationalization of the indifference curves: two-parameter subjective distributions	7
3.6	Second rationalization: quadratic utility	7
3.7	Effects of changes in the interest rate	8
3.8	Effects of changes in perceived risk and taxation	9
3.9	Multiple risky alternatives to cash	9
4	Identification and instruments (what makes the estimates credible)	10
4.1	What Tobin is trying to fix in the Keynesian theory	10
4.2	What assumptions generate the new result	11
4.3	Why diversification is central	11
4.4	Why the theory is useful but not all-powerful	11
4.5	Why the analysis is limited to monetary assets	11
5	Tables and figures	12

5.1	Figure 2.1, Figure 2.2, and Figure 2.3: the Keynesian sticky-expectations mechanism	12
5.2	Figure 3.1: opportunity locus and interior diversification	12
5.3	Figure 3.2 and Figure 3.3: corner solutions	12
5.4	Figure 3.4 and Figure 3.5: changes in risk and taxation	13
5.5	Figure 3.6: dominant combinations of risky assets	13
6	Short summary	14
7	Conclusion	14
7.1	Core conclusion (what the paper establishes)	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** Why do people hold cash even when cash pays no interest, and why should the aggregate demand for cash fall when the interest rate rises?
- **Setting.** The paper studies **liquidity preference** as a problem of **portfolio choice** among **monetary assets**: cash and interest-bearing government securities (“consols”). The focus is on the allocation *within* the class of monetary assets, not on the wider choice between money, equity, real estate, and other forms of wealth.
- **Core challenge.** The classic Keynesian explanation often relies on **inelastic expectations of future interest rates**: investors think rates will revert to a “normal” level, so low current rates imply expected capital losses on bonds and therefore a higher demand for cash. Tobin argues that this foundation is logically fragile, because if rates stay low long enough investors may revise what they regard as “normal.”
- **Main theoretical result.** Liquidity preference can instead be derived from **uncertainty** about future interest rates together with **risk aversion**. Even if the **expected capital gain on bonds is zero**, investors may still hold cash because cash is safer. In this formulation, a higher interest rate raises the reward for taking risk and can lower the desired cash share.
- **Why it matters.** The paper gives a more general and more behaviorally plausible foundation for downward-sloping money demand. It also connects monetary theory to modern portfolio choice: the investor chooses a point on a trade-off between **expected return** and **risk**.
- **What the paper changes relative to Keynes.**
 - It does *not* require expected capital losses on bonds.
 - It explains why the **same individual** can hold both cash and bonds.
 - It turns liquidity preference into a **mean–variance portfolio problem**.
- **Main analytical conclusions.**
 - With perfectly certain but sticky interest-rate expectations, each individual has an **all-or-nothing** portfolio rule: all cash or all consols depending on whether the current rate is below or above a critical threshold.
 - Aggregating across investors with different critical thresholds yields a familiar downward-sloping liquidity-preference schedule.
 - With uncertainty and risk aversion, interior portfolios naturally arise, so **diversification** is no longer anomalous.
 - For low interest rates, the theory implies a **negative elasticity of cash demand** with respect to the interest rate.
 - At high interest rates the sign can become ambiguous, although Tobin argues the standard inverse relationship is the empirically relevant case.
- **Economic takeaway.** Cash is not held because investors irrationally ignore yield. It is held because it is the **riskless pole** of the monetary-asset portfolio. Liquidity preference is therefore best understood as a **risk–return choice**, not simply as a bet on mean reversion in interest rates.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the “key objects” are

- There is **no paper-specific dataset**. The central objects are the asset categories, the return and risk measures attached to them, and the investor’s preference ordering over expected return and risk.
- The paper works in two steps:
 1. first, it restates the **Keynesian** story based on sticky expectations;
 2. then it develops Tobin’s **risk-aversion** alternative.

2.2 Transactions balances versus investment balances

- **Transactions balances.** These arise from imperfect synchronization between receipts and expenditures. Their aggregate magnitude is largely driven by institutional features of payment timing and the volume of transactions.
- **Composition of transactions balances.** Even transactions balances need not be held entirely in cash: sufficiently large transactors may temporarily place part of them in short-term earning assets.
- **Investment balances.** These are balances that will survive expected within-year spending needs. They do not need to be turned into cash quickly, so the composition of these balances is the genuine liquidity-preference problem.
- **Interpretation.** Tobin deliberately focuses on **investment balances**, because it is here that cash can only be justified by fears of capital loss on alternative monetary assets rather than by pure transactions convenience.

2.3 Assets, returns, and risk objects

- **Cash.** Cash has zero nominal yield and is taken to be the relatively **riskless** monetary asset.
- **Consols.** The benchmark interest-bearing asset is a perpetual government bond with current yield r . If the future yield changes, the consol experiences a capital gain or loss.
- **Portfolio shares.**
 - A_1 : share of the investment balance held in cash,
 - A_2 : share held in consols,
 - with $A_1 + A_2 = 1$.
- **Future capital gain/loss.** The random variable g measures the capital gain or loss per dollar invested in consols.
- **Expected portfolio return and risk.**
 - $\mu_R \equiv \mathbb{E}(R)$: expected return on the portfolio,

- σ_R : standard deviation of portfolio return,
- σ_g : perceived standard deviation of the capital gain/loss on consols.

2.4 Investor types, preferences, and timing

- The investor chooses the portfolio for a fixed decision period (a “year” in the exposition), recognizing that in reality transaction costs and information costs create inertia but not perfect rigidity.
- Tobin distinguishes **risk-lovers** from **risk-aversers**. The empirically relevant case is the risk-averter.
- Among risk-aversers, the important subtype is the **diversifier**: someone whose indifference curves are concave upward in (μ_R, σ_R) space and who therefore typically holds both cash and bonds.
- Tobin also discusses **plungers**: agents whose preferences may lead them to put everything at one corner (all cash or all consols). His constant-expected-utility derivations largely rule these agents out for the risk-averse case.

3 Models and empirical specifications (all equations + interpretation)

3.1 Keynesian benchmark: inelastic expectations of future interest rates

Tobin first reconstructs the classic speculative-demand argument. If the current consol yield is r and the investor expects the future yield to be r_e , then the capital gain or loss per dollar invested in consols is

$$g = \frac{r}{r_e} - 1. \quad (2.1)$$

If the investor compares cash with consols under certainty, the portfolio rule is all-or-nothing. There is a critical current rate

$$r_c = \frac{r_e}{1 + r_e}. \quad (2.2)$$

At current rates above r_c , the investor holds all consols; below r_c , all cash.

Interpretation.

- This is the classic Keynesian logic: if the current rate is low relative to the expected future “normal” rate, bonds are expected to suffer capital losses, so the investor stays in cash.
- The individual demand for cash is therefore a **step function**, not a smooth curve.
- The aggregate smooth liquidity-preference curve appears only after aggregating over many investors with different critical rates.

3.2 Aggregation under sticky expectations

If investors differ in their critical rates, aggregate cash demand becomes downward sloping even though each individual's demand is discontinuous.

Interpretation.

- At sufficiently high interest rates, nobody wants cash.
- At sufficiently low interest rates, everyone wants cash.
- In between, the total demand for cash equals the investment balances of those investors whose critical rates exceed the current rate.

Tobin also notes that if outstanding consols already exist, the **size** of investment balances changes with the interest rate because capital gains and losses alter wealth. That matters for the shape of the liquidity-preference schedule and for the interpretation of open-market operations.

3.3 The risk-aversion foundation: portfolio return, expected return, and risk

Tobin's alternative assumes that the investor is **uncertain** about the future interest rate and hence about the capital gain/loss g . He assumes, for the core argument, that the expected value of g is zero.

The return on the portfolio is

$$R = A_2(r + g), \quad 0 \leq A_2 \leq 1. \quad (3.1)$$

Expected return is

$$\mathbb{E}(R) = \mu_R = A_2 r. \quad (3.2)$$

Portfolio risk, measured by the standard deviation of return, is

$$\sigma_R = A_2 \sigma_g. \quad (3.3)$$

Combining these gives the investor's **opportunity locus** in (μ_R, σ_R) space:

$$\mu_R = \frac{r}{\sigma_g} \sigma_R, \quad 0 \leq \sigma_R \leq \sigma_g. \quad (3.4)$$

Interpretation.

- The more of the balance the investor puts into consols, the higher both expected return and risk.
- The slope r/σ_g is the reward for bearing one extra unit of interest-rate risk.
- A rise in r rotates the opportunity locus upward; a fall in perceived risk σ_g does the same.

3.4 Preferences over expected return and risk

Tobin assumes that the investor can rank portfolios by indifference curves in (μ_R, σ_R) space.

- A **risk-avorter** wants more expected return for any given risk and less risk for any given expected return.
- A **risk-lover** may accept lower expected return in exchange for more risk.
- The relevant case for liquidity preference is the risk-averse investor, whose indifference curves slope upward.

Interpretation. The portfolio problem is now: choose the point on the opportunity locus that touches the highest indifference curve.

3.5 First rationalization of the indifference curves: two-parameter subjective distributions

Suppose the subjective distribution of R is described by the two parameters (μ_R, σ_R) . Then expected utility is

$$\mathbb{E}[U(R)] = \int_{-\infty}^{\infty} U(R) f(R; \mu_R, \sigma_R) dR = \int_{-\infty}^{\infty} U(\mu_R + \sigma_R z) f(z; 0, 1) dz. \quad (3.5-3.6)$$

Holding expected utility constant, the slope of the indifference locus is

$$\frac{d\mu_R}{d\sigma_R} = - \frac{\int_{-\infty}^{\infty} z U'(\mu_R + \sigma_R z) f(z; 0, 1) dz}{\int_{-\infty}^{\infty} U'(\mu_R + \sigma_R z) f(z; 0, 1) dz}. \quad (3.7)$$

Interpretation.

- If marginal utility is positive and decreasing, the indifference curves slope upward and are concave upward.
- Therefore a risk-avorter is naturally a **diversifier**: the geometry itself favors interior portfolios.

3.6 Second rationalization: quadratic utility

Tobin then shows that the same mean–variance setup can be obtained if the utility-of-return function is quadratic:

$$U(R) = (1 + b)R + bR^2. \quad (3.8)$$

Here,

- $-1 < b < 0$ corresponds to a **risk-avorter**,
- $0 < b < 1$ corresponds to a **risk-lover**.

Under quadratic utility,

$$\mathbb{E}[U(R)] = (1 + b)\mu_R + b(\sigma_R^2 + \mu_R^2). \quad (3.12)$$

Holding expected utility constant yields the slope of an indifference curve:

$$\frac{d\mu_R}{d\sigma_R} = \frac{\sigma_R}{-\frac{1+b}{2b} - \mu_R}. \quad (3.13)$$

Interpretation.

- For a risk-avertter ($b < 0$), the denominator is positive over the relevant range, so the slope is positive.
- The second derivative is also positive, so the indifference curves are concave upward.
- This again makes the risk-avertter a **diversifier**.

3.7 Effects of changes in the interest rate

The investor chooses the highest indifference curve attainable on the opportunity locus. Tobin distinguishes three cases:

1. an interior tangency with both cash and consols positive;
2. a corner at all consols;
3. a corner at all cash.

Under quadratic utility, equating the indifference-curve slope and the opportunity-locus slope gives

$$\frac{r}{\sigma_g} = \frac{\sigma_R}{-\frac{1+b}{2b} - \mu_R}. \quad (3.15)$$

Differentiating the implied portfolio rule shows that the share of consols rises with the interest rate for sufficiently low rates and becomes less elastic at high rates.

Interpretation.

- The standard liquidity-preference result comes from the **substitution effect**: a higher yield on consols makes risk-bearing more attractive.
- But there is also an **income effect**: a higher rate lets the investor obtain more yield with less risk.
- Therefore the sign can be ambiguous at high rates.
- Tobin's important nuance is that the inverse cash–interest relationship is strongest at **low rates**; the theory is not committed to the same slope everywhere.

3.8 Effects of changes in perceived risk and taxation

A change in perceived bond risk has an even stronger effect on portfolio choice than an equal proportional change in the interest rate. Tobin summarizes the relationship between the elasticity of consol demand with respect to risk and with respect to the interest rate as

$$\frac{\sigma_g}{A_2} \frac{dA_2}{d\sigma_g} = -\frac{r}{A_2} \frac{dA_2}{dr} - 1. \quad (3.17)$$

Interpretation.

- If perceived interest-rate risk falls, investors move toward consols.
- This shift is stronger than an equal increase in the interest rate because both the opportunity locus and the risk–consol relation change.
- Central-bank actions that change perceived risk can therefore matter as much as direct open-market operations.

Tobin also studies taxation:

- A tax on both interest income and capital gains, with full loss offset, **reduces the demand for cash**.
- A tax on interest income alone, with no offset for capital losses, **increases the demand for cash**.

Interpretation. Taxes matter because they alter the mean-risk trade-off faced by investors, not just because they change average yield mechanically.

3.9 Multiple risky alternatives to cash

Tobin next allows for m non-cash monetary assets. Let x_i be the amount invested in risky asset i , with $\sum_{i=1}^m x_i = A_2 < 1$. Let r_i be the expected yield on asset i , and let g_i be its capital gain/loss.

The covariance matrix of risky-asset returns is

$$v_{ij} = \mathbb{E}(g_i g_j), \quad i, j = 1, \dots, m. \quad (3.18)$$

The portfolio's expected return is

$$\mu_R = \sum_{i=1}^m r_i x_i. \quad (3.19)$$

Its variance is

$$\sigma_R^2 = \sum_{i=1}^m \sum_{j=1}^m x_i x_j v_{ij}. \quad (3.20)$$

A **dominant combination** of risky assets minimizes variance for a given expected return:

$$\min_{x_i} \left\{ \sum_i \left(\sum_j v_{ij} x_j \right) x_i - \lambda \left(\sum_i r_i x_i - \mu_R \right) \right\}. \quad (3.21)$$

The first-order conditions are

$$[v_{ij}][x_i] = [\lambda r_i]. \quad (3.22)$$

If $[V_{ij}] = [v_{ij}]^{-1}$, then

$$\mu_R = \lambda \sum_i \sum_j r_i r_j V_{ij}, \quad (3.23)$$

$$\sigma_R^2 = \lambda^2 \sum_i \sum_j r_i r_j V_{ij}. \quad (3.24)$$

Hence the opportunity locus for the best risky-asset composite is again linear:

$$\mu_R = \sigma_R \sqrt{\sum_i \sum_j r_i r_j V_{ij}}. \quad (3.25)$$

Interpretation.

- All dominant risky-asset portfolios lie on a **ray from the origin**.
- The investor can therefore solve the problem in two stages:
 1. choose the **efficient composite** of risky monetary assets;
 2. choose how much to allocate between that composite and **cash**.
- Cash is crucial here because it is the residual **riskless** asset. Without a riskless asset, the investor's feasible locus is generally nonlinear and the clean two-stage simplification disappears.

4 Identification and instruments (what makes the estimates credible)

4.1 What Tobin is trying to fix in the Keynesian theory

- The sticky-expectations theory can generate a downward-sloping aggregate liquidity-preference schedule.
- But its individual-level logic is brittle:
 - each investor chooses **all cash or all bonds**,
 - the key driver is a fixed “normal” rate of interest,
 - critics such as Leontief and Fellner argue that such a foundation is hard to reconcile with equilibrium reasoning.

Tobin's move is to replace this with a theory in which the relevant friction is not stubborn mean reversion per se, but **uncertainty about future rates** combined with **risk aversion**.

4.2 What assumptions generate the new result

- Investors face uncertainty about the capital gain/loss on interest-bearing assets.
- The expected capital gain can be set to zero, so the argument does not rely on systematic expectations of losses.
- Investors rank portfolios by expected return and risk.
- Risk-averse investors have concave-upward indifference curves and thus generally diversify between cash and consols.

Interpretation. Once these assumptions are imposed, liquidity preference follows from the standard trade-off between yield and safety.

4.3 Why diversification is central

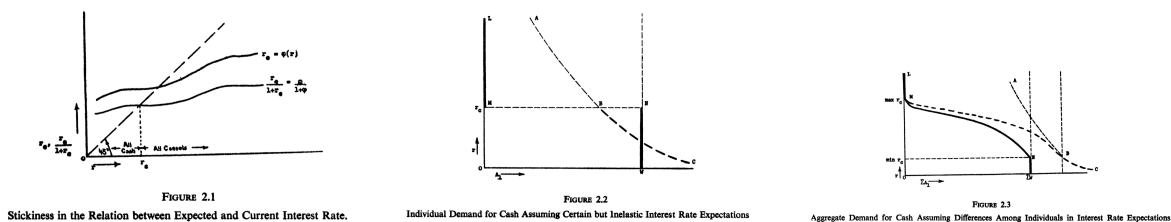
- The Keynesian version explains aggregate money demand but does poorly at the individual level because it predicts corner solutions.
- Tobin's theory explains why the **same person** may hold both cash and bonds.
- This is empirically attractive because most real-world investors are diversified at least to some extent.

4.4 Why the theory is useful but not all-powerful

- The theory works best for **low interest rates**, where the inverse relation between cash demand and the interest rate is most robust.
- At high rates, income and substitution effects may conflict, so cash demand can in principle rise rather than fall with the interest rate.
- In a strict stationary state, Tobin concedes that liquidity preference may disappear because the perceived risk of capital loss would eventually vanish if rates had been constant for long enough.

4.5 Why the analysis is limited to monetary assets

- Tobin explicitly restricts the analysis to **cash and other monetary assets**.
- In a broader portfolio problem, cash is not literally riskless because all monetary assets are exposed to inflation and because equities and real assets introduce different forms of risk.
- The paper therefore advocates a staged simplification: first choose across broad asset classes, then choose within the class of monetary assets.



5 Tables and figures

5.1 Figure 2.1, Figure 2.2, and Figure 2.3: the Keynesian sticky-expectations mechanism

Read:

- Figure 2.1 illustrates how a sticky mapping from the current rate r to the expected future rate r_e creates a **critical current rate** r_c .
- Figure 2.2 shows the individual implication: cash demand is a **step function**. One investor switches discontinuously from all cash to all consols.
- Figure 2.3 shows how aggregation across heterogeneous critical rates smooths these step functions into a conventional downward-sloping liquidity-preference curve.
- This trio of figures is important because it states the benchmark theory that Tobin is about to replace.

5.2 Figure 3.1: opportunity locus and interior diversification

Read:

- The upper panel maps the trade-off between expected return and risk; the lower panel maps that same choice into holdings of cash and consols.
- Rays OC_1 , OC_2 , and OC_3 represent different interest rates.
- Tangency points T_1 , T_2 , and T_3 illustrate the diversified interior solution for a risk-avertter.
- This is the main figure of the paper: cash demand is a portfolio choice, not an all-or-nothing speculative bet.

5.3 Figure 3.2 and Figure 3.3: corner solutions

Read:

- Figure 3.2 shows the all-consols corner. This can occur for a risk-lover or for a risk-avertter whose preferences still put the optimum at the maximum-risk point.
- Figure 3.3 shows the possibility of all-cash or all-consols corner solutions for Tobin's "plungers."
- The crucial point is that Tobin's constant-expected-utility derivations largely exclude plungers for the risk-averse case. Hence diversified portfolios are the normal prediction under risk aversion.

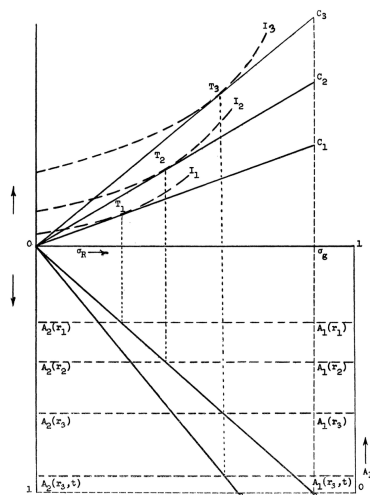


FIGURE 3.1
Portfolio Selection at Various Interest Rates and Before and After Taxation.

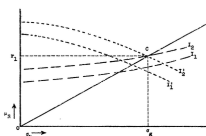


FIGURE 3.2
"Risk-lovers" and "Diversifiers": Optimum Portfolio at Maximum Risk and Expected Return.

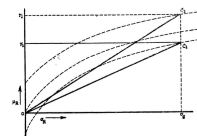


FIGURE 3.3
"Plungers"—Optimum Portfolio at Minimum or Maximum Risk and Expected Return.

5.4 Figure 3.4 and Figure 3.5: changes in risk and taxation

Read:

- Figure 3.4 compares a change in the interest rate with a change in perceived risk.
- Tobin's key message is that lowering perceived risk can shift the investor more strongly toward consols than an equally proportioned increase in the interest rate.
- Figure 3.5 then translates this geometry into a liquidity-preference schedule under taxation.
- Equal taxation of interest and capital gains with loss offsets shifts the schedule toward **less cash**; taxing interest only shifts it toward **more cash**.

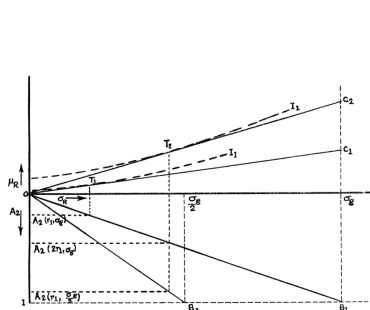


FIGURE 3.4
Comparison of effects of changes in interest rate (r) and in "risk" (σ_p) on holding of consols.

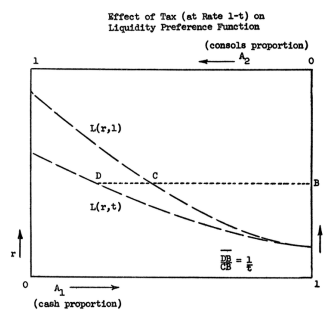


FIGURE 3.5
Effect of Tax (at Rate $1-t$) on Liquidity Preference Function.

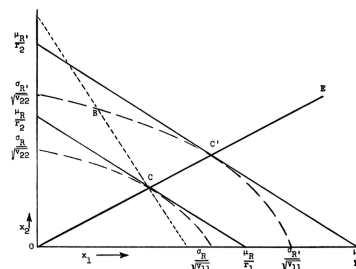


FIGURE 3.6
Dominant Combinations of Two Assets.

5.5 Figure 3.6: dominant combinations of risky assets

Read:

- The figure draws constant-return lines and constant-risk ellipses for two risky assets.
- Points such as C and C' are **dominant** combinations: for a given risk they maximize expected return, or for a given expected return they minimize risk.

- All such dominant combinations lie on a **ray**, which is the geometric core of the two-stage portfolio result.

6 Short summary

- **Method.** Reinterpret liquidity preference as a mean–variance portfolio problem over monetary assets.
- **Benchmark.** The Keynesian explanation based on sticky expectations yields all-or-nothing portfolio choices and is logically fragile.
- **Main result.** With uncertainty about future rates and risk aversion, investors hold cash because it is safer, and a higher interest rate generally reduces desired cash balances, especially at low rates.
- **Additional result.** With many risky monetary assets, the investor first chooses an efficient risky composite and then chooses how much cash to hold against that composite.
- **Bottom line.** Liquidity preference is best seen as behavior toward risk.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Tobin shows that the demand for cash does not need to be explained by expected capital losses on bonds.
- It can instead arise from **risk aversion under uncertainty**.
- This gives a more satisfactory foundation for liquidity preference and better explains diversified monetary portfolios.

7.2 Economic intuition

- Cash yields little, but it is safe.
- Bonds yield more, but their market value fluctuates when interest rates move.
- Investors therefore trade off yield against safety.
- The demand for money is one part of a broader portfolio problem rather than a separate behavioral anomaly.

7.3 Takeaway

This paper is important because it rewrites a Keynesian monetary concept in the language of portfolio theory. The central lesson is simple: people hold money not because they fail to see the yield on other assets, but because money sits at the safe end of the risk–return frontier. That is the foundation of liquidity preference that Tobin wants macroeconomics to use.